Chaos Theory

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1 Abstract

Chaos theory is the study of nonlinear dynamics, where seemingly random events are actually predictable from simple deterministic equations. In Chaos theory, one studies the behavior of dynamical systems that are highly sensitive to initial conditions, an effect which is popularly referred to as the **butterfly effect**. The two main components of chaos theory are the ideas that systems - no matter how complex rely upon an underlying order, and that very simple or small systems and events can cause very complex behaviors or events.

In this project I have studied chaos through multiple demographic models :
1. Logistic Map
2. Sine map
3. Dyadic Map
4. Double pendulum or Chaotic Pendulum.

In particular, a chaotic dynamical system is generally characterized by
1. Being sensitive to the initial condition of the system (so that initially nearby points can evolve quickly into very different states).
2. A dense set of unstable periodic orbits.
3. Some trajectory orbits being aperiodic, meaning that they do not repeat themselves on any time scales.
4. Mixing

1.1 Basic Definitions for understanding the behavior of One dimensional maps

• **Fixed point**:
Fixed are very useful to understand the dynamical behavior of one dimensional maps. In one dimensional map a point is said to be a fixed point $\dot{x}$ when
f($\dot{x}$) = $\dot{x}$
For a stable fixed point trajectories of the points near the fixed point must converge to fixed point. So let us suppose $\dot{x}$ to be a fixed point and x_n be a point nearby. Therefore for a stable fixed points
$\delta x_{n+1} \leq \delta x_n$
• **Bifurcation**
Bifurcation deals with sudden change in dynamics of the system.In the given context the change in the character of fixed points due to changes in parameters of the system will be referred as bifurcation. In case of uni-modal quadratic maps pitchfork bifurcation takes place. Bifurcation diagram of a given map is the graph between the parameter used against the asymptotic value of the independent variable.
• **Lyapunov Exponent**
Lyapunov exponent is one of the ways to measure the divergence at nearby trajectories. For a dynamical system if initial separation between trajectories is x_0 the at a later time t let the separation between trajectories be s then:
$s_t = s_0 e^{\lambda t}$ where λ is known as Lyapunov exponent.For chaotic trajectories $\lambda \geq 0$ is a necessary condition. In this case the nearby trajectories will diverge away exponentially which will make it unpredictable.

2 Logistic Map

The Logistic map,

$$x_{n+1} = \alpha x_n(1 - x_n) \tag{1}$$

where α is a constant.
is an used as an example to explain how simple non-linear functions can help in demonstrating chaos.
It is a second order non-linear recurrence relation.
Observing the nonlinear function one can see that if we take either large positive or negative values of x_n the series rapidly diverges to -∞
So to observe the behavior of the function we restrict x_n between 0 and 1 and α between 0 and 4 and divide the interval into several subintervals to observe different behaviour.

2.1 C++ Program for Logistic Map Analysis

The following C++ program produces a random number between 0 and 4 for α and a random number between 0 and 1 for x_n.

hyperref showspaces

```
/*
 x(n+1)=a*x(n)[1-x(n)]
0<a<4
0<x<1
*/
#include<iostream>
#include<math.h>
#include<stdlib.h>
#include<time.h>
#include<fstream>
using namespace std;
int main()
{
srand (time(NULL));
ofstream fout("alpha.txt");
float a,x0,x1;
int n =1 + rand()%400;
a=(float)n/100;
int k=1+rand()
%100;
x0=(float)k/100;


cout<<"alpha="<<a;
cout<<"\nx0="<<x0<<endl;
fout<<"alpha="<<a<<endl;
fout<<"0"<<"\t"<<x0<<"\n";
for(int i=1;i<=100;i++)
{
x1=a*x0*(1-x0);
fout<<i<<"\t"<<x1<<"\n";
x0=x1;
}


fout.close();
return 0;
}
```

For the logistic map equation the fixed point is :

$$x_{n+1} = \alpha x_n(1 - x_n) \tag{2}$$

$$1 = \alpha(1 - x_n) \tag{3}$$

i.e.

$$x_n = \frac{\alpha - 1}{\alpha} \tag{4}$$

Observations:-

After running the above program for distinct input values the following observations are made:

1. $\alpha \in (0, 1)$ then the series rapidly diverges to zero.

2. $\alpha \in (1, 2)$ then the series rapidly converges to a value equal to $\frac{\alpha-1}{\alpha}$ for any value of $x_n \in (0, 1)$

3. $\alpha \in (2, 3)$ then the series converges to the same point i.e. $\frac{\alpha-1}{\alpha}$ but with a slower rate which is still linear.

4. $\alpha \in (3, 3.45)$ the series oscillates between two points which depend on α.

5. $\alpha \in (3.45, 3.54)$ the series oscillates between four points which also depend on α

The number of points for oscillations keeps on increasing as 8,16,32 and so on.

With α near 3.57 is the onset of chaos where even small changes in intial values give rise to exponential differences in final values.

2.2 Mathematical Analysis

We have to check the stability of the fixed point $\dot{x}$ (say).
To check the stability of the fixed point we displace it by a small amount δ_0 (say) and see if the series has a tendency to bring the point back to $\dot{x}$.

2.2.1 $\alpha \in [0, 1]$

We have,
1)

$$x_n = \dot{x} + \delta_0 \tag{5}$$

2)

$$x_{n+1} = \dot{x} + \delta_1 \tag{6}$$

and

$$\begin{aligned} x_{n+1} &= \alpha x_n(1 - x_n) \\ &= \alpha[x_n - x_n^2] \\ &= \alpha[\dot{x} + \delta_0 - (\dot{x} + \delta_0)^2] \\ &= \alpha[\dot{x} + \delta_0 - (\dot{x}^2 + \delta_0^2 + 2\dot{x}\delta_0)] \end{aligned}$$

Since δ_0 is very small implies $\delta_0^2 << 1$ and can be neglected.

$$= \alpha[\dot{x} + \delta_0 - \dot{x}^2 - 2\dot{x}\delta_0]$$

$$i.e, \dot{x} + \delta_1 = \alpha[\dot{x} + \delta_0 - \dot{x}^2 - 2\dot{x}\delta_0]$$

Put $\dot{x} = 0$ we get

$$\delta_1 = \alpha * \delta_0$$

And since $\alpha \in (0, 1)$
Therefore, $\delta_1 < \delta_0$
i.e. series converges and hence the fixed point namely $\dot{x} = 0$ is stable.

2.2.2 $\alpha \in (1, 2]$

$$\dot{x} + \delta_1 = \alpha[\dot{x}(1 - \dot{x}) + \delta_0(1 - 2\dot{x})] \tag{7}$$

We have $\dot{x} = \frac{\alpha-1}{\alpha}$

$$\begin{aligned} i.e, \frac{\alpha - 1}{\alpha} + \delta_1 &= \alpha[\frac{\alpha - 1}{\alpha} * \frac{1}{\alpha} + \delta_0 \frac{2 - \alpha}{\alpha}] \\ i.e, \frac{\alpha - 1}{\alpha} + \delta_1 &= \frac{\alpha - 1}{\alpha} + \delta_0(2 - \alpha) \\ i.e, \delta_1 &= \delta_0(2 - \alpha) \end{aligned} \tag{8}$$

Since $\alpha \in (1, 2]$,therefore $2 - \alpha \in [0, 1)$
Therefore, $\delta_1 < \delta_0$
i.e. series converges to $\dot{x} = \frac{\alpha-1}{\alpha}$ and is a stable fixed point.

2.2.3 $\alpha \in (2, 3)$

Since $\alpha \in (2, 3)$,therefore $|2 - \alpha| \in (0, 1)$
Substituting in equation 8 we see:
$|\delta_1| < |\delta_0|$
i.e. series converges to $\dot{x} = \frac{\alpha-1}{\alpha}$ and is a stable fixed point.
Although the rate of convergence is slower than previous case.

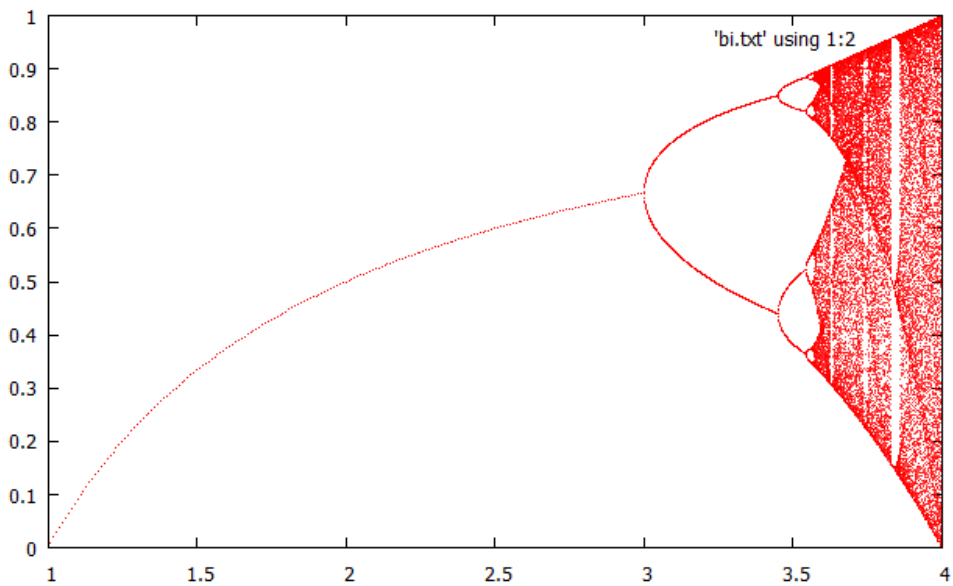
2.2.4 $\alpha \in (3, 4)$

Since $\alpha \in (3, 4)$,therefore $|2 - \alpha| \in (1, 2)$
Substituting in equation 8 we see:
$|\delta_1| > |\delta_0|$
i.e. series diverges and their is no stable fixed point.

Finally after α is greater than 3.57 , the prime characteristic of chaos can be observed i.e. exponential differences in results for minute changes in initial conditions.

2.3 Bifurcation Diagram

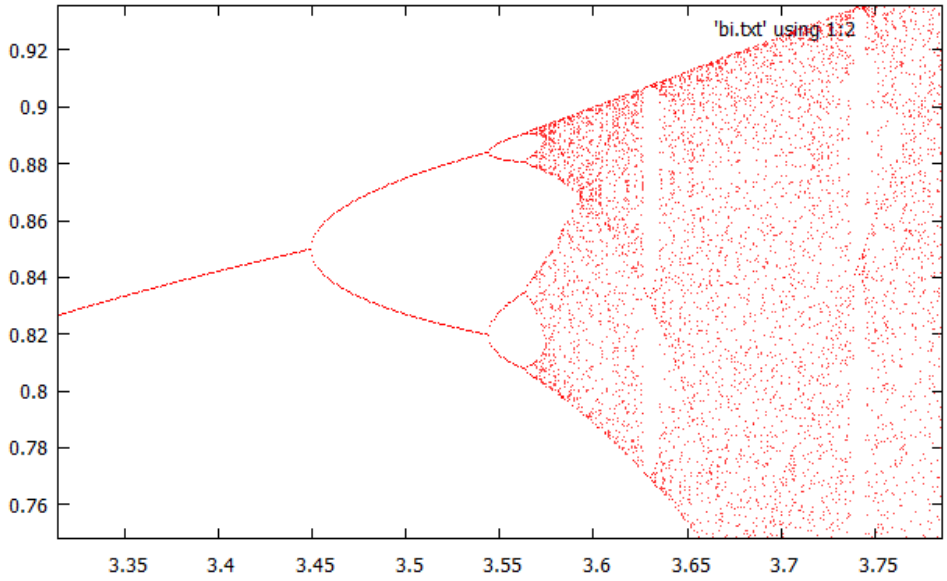
A bifucation is a period-doubling, a change from an N-point attractor to a 2N-point attractor, which occurs when the control parameter is changed.
A Bifurcation Diagram is a visual summary of the succession of period-doubling produced as α increases. The next figure shows the bifurcation diagram of the logistic map, α along the x-axis. For each value of α the system is first allowed to settle down and then the successive values of x are plotted for a few hundred iterations.



We see that for α less than one, all the points are plotted at zero. Zero is the one point attractor for α less than one. For α between 1 and 3, we still have one-point attractors, but the 'attracted' value of x increases as α increases, at least to α=3. Bifurcations occur at α=3, α=3.45, 3.54, 3.564, 3.569 (approximately), etc., until just beyond 3.57, where the system is chaotic.
However, the system is not chaotic for all values of α greater than 3.57.

Notice that at several values of α, greater than 3.57, a small number of x values are visited. These regions produce the 'white space' in the diagram. Look closely at α=3.83 and you will see a three-point attractor.
In fact, between 3.57 and 4 there is a rich interleaving of chaos and order. A small change in α can make a stable system chaotic, and vice versa.

We have illustrated here one of the symptoms of chaos. A chaotic system is one for which the distance between two trajectories from nearby points in its state space diverge over time. The magnitude of the divergence increases exponentially in a chaotic system.



Here is an image of the same bifurcation map as above zoomed in further at $\alpha = 3.35$. We see that the zoomed image is same as the original one. This self symmetry of the bifurcation diagram goes on further and further and due to this self symmetry in Logistic Map is the presence of a universal constant known as Feigenbaum constant, $\delta = 4.6692016$

3 Sine Map

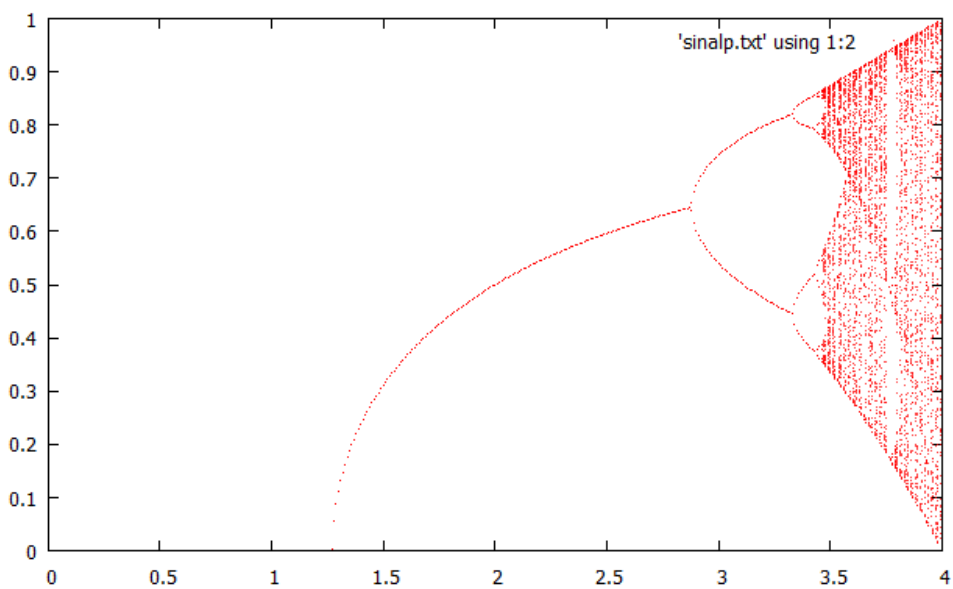
Sine map is another 1D map similar to the Logistic Map given as:

$$F(x) = \frac{\alpha}{4}\sin(\pi x_n) \tag{9}$$

where α is the parameter varied from 0 to 4 to observe the charactristics of the map.

Continuous function corresponding to this discrete map will be:
$f(x) = \alpha \sin(\pi x)$
This function also has only one maxima in the interval [0:1] (i. e. at x=0.5) and to map interval [0:1] back to same interval [0:1] we will choose $\alpha \in [0 : 1]$
Therefore this map is also a uni-modal quadratic map and its behavior will be similar to logistic map.

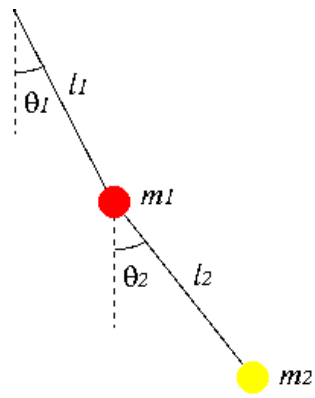
Bifurcation diagram for sin map is similar to that of logistic map. Initially as only stable fixed point is 0, so bifurcation diagram shows zero while after $\alpha = 0.3184$ it becomes unstable and tends to one fixed point. At $\alpha = 0.7200$(approx.) second bifurcation takes place and period one becomes unstable and period two starts by pitchfork bifurcation. Similarly other periods are observed at different bifurcation points via period doubling. And at $_\infty$ = infinite period is observed which is onset of chaos and bifurcation diagram becomes dense. After that point sin map becomes chaotic ,however it is disturbed by period which starts at α= 0.938. By knowing the value of A1 , A2 the parameter value for infinite period can be obtained by using Feigenbaum constant. Here A1 = 0.71992301,A2 = 0.83324980,δ = 4.6692016, therefore- $A_\infty = 0.857520928$



Bifurcation diagram for sin map for $x_0 = 0.4$, from $\alpha \in [0 : 4]$ in 20000 iterations and with an interval of 0.0001

4 Double Pendulum Dynamics

A planar double pendulum is a simple mechanical system that has two simple pendula attached end to end that exhibits chaotic behavior. First, the physical system is introduced and a system of coordinates is fixed, and then the Lagrangian and the Hamiltonian equations of motions are derived. We will find that the system is governed by a set of coupled nonlinear ordinary differential equations and using these, the system can be simulated. Finally we analyze Poincare sections, the largest lyapunov exponent, progression of trajectories, and change of angular velocities with time for certain system parameters at varying initial conditions.



Consider a double bob pendulum with masses and attached by rigid massless wires of lengths l_1 and l_2 . Further, let the angles the two wires make with the vertical be denoted θ_1 and θ_2 , as illustrated above. Finally, let gravity be given by g. Then the positions of the bobs are given by :

$$x_1 = l_1 \sin(\theta_1) \tag{10}$$

$$y_1 = -l_1 \cos(\theta_1) \tag{11}$$

$$x_2 = l_1 \sin(\theta_1) + l_2 \sin(\theta_2) \tag{12}$$

$$y_2 = -l_1 \cos(\theta_1) - l_2 \cos(\theta_2) \tag{13}$$

The potetial energy of the system is given by:

$$V = m_1 g y_1 + m_2 g y_2 \tag{14}$$

$$= -(m_1 + m_2) g l_1 \cos(\theta_1) - m_2 g l_2 \cos(\theta_2) \tag{15}$$

and the kinetic energy by:

$$T = \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2 \tag{16}$$

$$T = \frac{1}{2} m_1 l_1^2 \dot{\theta_1}^2 + \frac{1}{2} m_2 [l_1^2 \dot{\theta_1}^2 + l_2^2 \dot{\theta_2}^2 + 2 l_1 l_2 \dot{\theta_1} \dot{\theta_2} \cos(\theta_1 - \theta_2)] \tag{17}$$

The Lagrangian is then:

$$L = T - V \tag{18}$$

$$= \frac{1}{2} m_1 l_1^2 \dot{\theta_1}^2 + \frac{1}{2} m_2 [l_1^2 \dot{\theta_1}^2 + l_2^2 \dot{\theta_2}^2 + 2 l_1 l_2 \dot{\theta_1} \dot{\theta_2} \cos(\theta_1 - \theta_2)] + (m_1 + m_2) g l_1 \cos(\theta_1) + m_2 g l_2 \cos(\theta_2) \tag{19}$$

Therefore for θ_1

$$\frac{\partial L}{\partial \dot{\theta_1}} = p_{\theta_1} = m_1 l_1^2 \dot{\theta}_1 + m_2[l_1^2\dot{\theta}_1 + l_1 l_2 \dot{\theta}_2 \cos(\theta_1 - \theta_2)] \tag{20}$$

$$\frac{\partial L}{\partial \dot{\theta_1}} = p_{\theta_2} = m_2[l_2^2\dot{\theta}_2 + l_1 2\dot{\theta}_1 \cos(\theta_1 - \theta_2)] \tag{21}$$

And for θ_2

$$\frac{\partial L}{\partial \theta_1} = \dot{p}_{\theta_1} = m_2 l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 \sin(\theta_1 - \theta_2) - m_1 g l_1 \sin(\theta_1) - m_2 g l_1 \sin(\theta_1) \tag{22}$$

$$\frac{\partial L}{\partial \theta_2} = \dot{p}_{\theta_2} = -m_2 l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 \sin(\theta_1 - \theta_2) - m_2 g l_2 \sin(\theta_2) \tag{23}$$

The Hamiltonian is then given by: $H = \sum \theta_i p_i - L$

$$H = \frac{m_2 l^2 p_{\theta_1}^2 + l_1^2(m_1 + m_2)p_{\theta_2}^2 - 2m_2 l_1 l_2 p_{\theta_1} p_{\theta_2} \cos(\theta_1 - \theta_2)}{2l_1^2 l_2^2 m_2(m_1 + m_2 \sin(\theta_1 - \theta_2)^2)} \tag{24}$$

And Hamilton's equations of motion are given by:

$$\dot{\theta}_1 = \frac{\partial H}{\partial p_{\theta_1}} = \frac{l_2 p_{\theta_1} - l_1 p_{\theta_2} \cos(\theta_1 - \theta_2)}{l_1^2 l_2[m_1 + m_2 \sin(\theta_1 - \theta_2)^2]} \tag{25}$$

$$\dot{\theta}_2 = \frac{\partial H}{\partial p_{\theta_2}} = \frac{l_1(m_1 + m_2)p_{\theta_2} - l_2 m_2 p_{\theta_1} \cos(\theta_1 - \theta_2)}{l_1 l_2^2[m_1 + m_2 \sin(\theta_1 - \theta_2)^2]} \tag{26}$$

$$\dot{p_{\theta_1}} = -\frac{\partial H}{\partial \theta_1} = -(m_1 m_2) g l_1 \sin(\theta_1) - C_1 + C_2 \tag{27}$$

$$\dot{p_{\theta_2}} = -\frac{\partial H}{\partial \theta_2} = -m_2 g l_2 \sin(\theta_2) + C_1 - C_2 \tag{28}$$

where

$$C_1 = \frac{p_{\theta_1} p_{\theta_2} \sin(\theta_1 - \theta_2)}{l_1 l_2[m_1 + m_2 \sin(\theta_1 - \theta_2)^2]} \tag{29}$$

$$C_2 = \frac{l_2 m_2 p_1^2 + l_1^2(m_1 + m_2)p_2^2 - l_1 l_2 m_2 p_1 p_2 \cos(\theta_1 - \theta_2)}{2l_1^2 l_2^2[m_1 + m_2 \sin(\theta_1 - \theta_2)^2]^2} \sin[2(\theta_1 - \theta_2)] \tag{30}$$

Simulating these equations show the chaotic dynamics of a double pendulum.

5 Conclusion

Through a nice and interesting study of Lagrangian and Hamiltonian mechanics and with a detailed study of one dimensional maps I was able to understand the most basic of all questions i.e. what is the difference between chaos and randomness.
Along with this we had a basic introduction in Quantum Mechanics with the bra ket algebra to the various finite square well
problems and delta function potentials.We also studied even and odd functions and the fact that confinement of any form leads
to quantisation of some physical property.

6 References

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7 Acknowledgements

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